

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Enterprise Resource Planning System</title>
```

```
<style>
```

```
/* ERP SYSTEM */
```

```
:root {
  --primary: #2563eb;
  --primary-dark: #1d4ed8;
  --bg: #f4f7fb;
  --card: #ffffff;
  --text: #1e293b;
  --muted: #64748b;
  --border: #e2e8f0;
  --sidebar: #0f172a;
  --sidebar-text: #cbd5e1;
  --sidebar-hover: #1e293b;
  --success: #16a34a;
  --danger: #dc2626;
  --warning: #d97706;
  --info: #0284c7;
  --input: #ffffff;
  --shadow: 0 4px 18px rgba(0,0,0,0.08);
}
```

```
[data-theme="dark"] {
  --primary: #60a5fa;
```

```
--primary-dark: #3b82f6;
--bg: #0f172a;
--card: #1e293b;
--text: #f1f5f9;
--muted: #94a3b8;
--border: #334155;
--sidebar: #020617;
--sidebar-text: #cbd5e1;
--sidebar-hover: #1e293b;
--input: #0f172a;
--shadow: 0 4px 20px rgba(0,0,0,0.35);
}

* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: Arial, Helvetica, sans-serif;
  background: var(--bg);
  color: var(--text);
  transition: background 0.25s, color 0.25s;
}

button,
input,
select {
  font-family: inherit;
}
```

```
button {  
    cursor: pointer;  
}
```

```
/* =====  
LOGIN / REGISTER  
===== */
```

```
.auth-container {  
    min-height: 100vh;  
    display: flex;  
    align-items: center;  
    justify-content: center;  
    background:  
        radial-gradient(circle at top left, rgba(37,99,235,0.25), transparent 35%),  
        radial-gradient(circle at bottom right, rgba(14,165,233,0.2), transparent 35%),  
        var(--bg);  
    padding: 20px;  
}
```

```
.auth-box {  
    width: 100%;  
    max-width: 440px;  
    background: var(--card);  
    border: 1px solid var(--border);  
    border-radius: 18px;  
    padding: 35px;  
    box-shadow: var(--shadow);  
}
```

```
.logo {  
  width: 70px;  
  height: 70px;  
  margin: auto;  
  background: var(--primary);  
  color: white;  
  border-radius: 18px;  
  display: flex;  
  align-items: center;  
  justify-content: center;  
  font-size: 28px;  
  font-weight: bold;  
}
```

```
.auth-box h1 {  
  text-align: center;  
  margin-top: 18px;  
}
```

```
.auth-subtitle {  
  text-align: center;  
  color: var(--muted);  
  margin: 8px 0 25px;  
}
```

```
.form-group {  
  margin-bottom: 15px;  
}
```

```
.form-group label {  
  display: block;
```

```
margin-bottom: 7px;  
font-weight: 600;  
}
```

```
input,  
select {  
  width: 100%;  
  padding: 12px 13px;  
  border: 1px solid var(--border);  
  border-radius: 8px;  
  background: var(--input);  
  color: var(--text);  
  outline: none;  
}
```

```
input:focus,  
select:focus {  
  border-color: var(--primary);  
}
```

```
.btn {  
  border: none;  
  padding: 11px 17px;  
  border-radius: 8px;  
  font-weight: 600;  
  transition: 0.2s;  
}
```

```
.btn-primary {  
  background: var(--primary);  
  color: white;
```

```
}
```

```
.btn-primary:hover {  
  background: var(--primary-dark);  
}
```

```
.btn-success {  
  background: var(--success);  
  color: white;  
}
```

```
.btn-danger {  
  background: var(--danger);  
  color: white;  
}
```

```
.btn-warning {  
  background: var(--warning);  
  color: white;  
}
```

```
.btn-secondary {  
  background: #64748b;  
  color: white;  
}
```

```
.btn-full {  
  width: 100%;  
}
```

```
.auth-switch {
```

```
text-align: center;
margin-top: 22px;
color: var(--muted);
}
```

```
.auth-switch button {
background: none;
border: none;
color: var(--primary);
font-weight: bold;
}
```

```
.demo-login {
margin-top: 20px;
padding: 12px;
border-radius: 8px;
background: rgba(37,99,235,0.1);
color: var(--text);
font-size: 13px;
}
```

```
.password-note {
color: var(--muted);
font-size: 12px;
margin-top: 5px;
}
```

```
/* =====
APP
===== */
```

```
#app {  
  display: none;  
  min-height: 100vh;  
}  
  
.layout {  
  display: flex;  
  min-height: 100vh;  
}  
  
.sidebar {  
  width: 250px;  
  background: var(--sidebar);  
  color: var(--sidebar-text);  
  position: fixed;  
  left: 0;  
  top: 0;  
  bottom: 0;  
  overflow-y: auto;  
  z-index: 100;  
}  
  
.brand {  
  padding: 22px;  
  font-size: 20px;  
  font-weight: bold;  
  color: white;  
  border-bottom: 1px solid rgba(255,255,255,0.08);  
}  
  
.brand span {
```

```
    color: #60a5fa;
}
```

```
.user-info {
    padding: 18px;
    border-bottom: 1px solid rgba(255,255,255,0.08);
}
```

```
.user-avatar {
    width: 42px;
    height: 42px;
    background: var(--primary);
    color: white;
    border-radius: 50%;
    display: flex;
    align-items: center;
    justify-content: center;
    font-weight: bold;
    float: left;
    margin-right: 12px;
}
```

```
.user-name {
    font-weight: bold;
    color: white;
}
```

```
.user-role {
    color: #94a3b8;
    font-size: 12px;
    margin-top: 4px;
```

```
}
```

```
.nav {  
  padding: 12px;  
}
```

```
.nav button {  
  width: 100%;  
  text-align: left;  
  background: transparent;  
  color: var(--sidebar-text);  
  border: none;  
  padding: 12px;  
  border-radius: 8px;  
  margin-bottom: 4px;  
  font-size: 14px;  
}
```

```
.nav button:hover,  
.nav button.active {  
  background: var(--sidebar-hover);  
  color: white;  
}
```

```
.nav-section {  
  color: #64748b;  
  font-size: 11px;  
  text-transform: uppercase;  
  margin: 20px 10px 8px;  
}
```

```
.sidebar-bottom {  
  padding: 12px;  
  position: sticky;  
  bottom: 0;  
  background: var(--sidebar);  
}
```

```
.main {  
  margin-left: 250px;  
  width: calc(100% - 250px);  
}
```

```
.topbar {  
  height: 70px;  
  background: var(--card);  
  border-bottom: 1px solid var(--border);  
  display: flex;  
  align-items: center;  
  justify-content: space-between;  
  padding: 0 25px;  
  position: sticky;  
  top: 0;  
  z-index: 50;  
}
```

```
.page-title {  
  font-size: 20px;  
  font-weight: bold;  
}
```

```
.top-actions {
```

```
    display: flex;
    gap: 10px;
    align-items: center;
}
```

```
.theme-toggle {
    border: 1px solid var(--border);
    background: var(--card);
    color: var(--text);
    border-radius: 20px;
    padding: 8px 14px;
}
```

```
.content {
    padding: 25px;
}
```

```
/* =====
CARDS
===== */
```

```
.cards {
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 18px;
    margin-bottom: 25px;
}
```

```
.stat-card {
    background: var(--card);
    border: 1px solid var(--border);
```

```
padding: 20px;

border-radius: 12px;

box-shadow: var(--shadow);
}
```

```
.stat-card .icon {

  font-size: 25px;
}
```

```
.stat-title {

  color: var(--muted);

  font-size: 13px;

  margin-top: 12px;
}
```

```
.stat-value {

  font-size: 27px;

  font-weight: bold;

  margin-top: 5px;
}
```

```
.section {

  background: var(--card);

  border: 1px solid var(--border);

  border-radius: 12px;

  padding: 20px;

  margin-bottom: 20px;

  box-shadow: var(--shadow);
}
```

```
.section-header {
```

```
display: flex;
justify-content: space-between;
align-items: center;
margin-bottom: 18px;
gap: 10px;
}
```

```
.section-header h2 {
  font-size: 18px;
}
```

```
/* =====
TABLE
===== */
```

```
.table-container {
  overflow-x: auto;
}
```

```
table {
  width: 100%;
  border-collapse: collapse;
}
```

```
th,
td {
  padding: 12px;
  text-align: left;
  border-bottom: 1px solid var(--border);
  font-size: 14px;
}
```

```
th {  
  color: var(--muted);  
  font-size: 12px;  
  text-transform: uppercase;  
}
```

```
tr:hover {  
  background: rgba(148,163,184,0.05);  
}
```

```
.badge {  
  display: inline-block;  
  padding: 5px 9px;  
  border-radius: 20px;  
  font-size: 11px;  
  font-weight: bold;  
}
```

```
.badge-success {  
  background: rgba(22,163,74,0.15);  
  color: var(--success);  
}
```

```
.badge-danger {  
  background: rgba(220,38,38,0.15);  
  color: var(--danger);  
}
```

```
.badge-warning {  
  background: rgba(217,119,6,0.15);
```

```
    color: var(--warning);
}
```

```
.badge-info {
    background: rgba(2,132,199,0.15);
    color: var(--info);
}
```

```
/* =====
FORMS
===== */
```

```
.form-grid {
    display: grid;
    grid-template-columns: repeat(3, 1fr);
    gap: 15px;
}
```

```
.form-actions {
    margin-top: 10px;
    display: flex;
    gap: 10px;
}
```

```
/* =====
REPORTS
===== */
```

```
.progress {
    height: 9px;
    background: var(--border);
}
```

```
border-radius: 10px;
overflow: hidden;
}
```

```
.progress-bar {
  height: 100%;
  background: var(--primary);
}
```

```
.report-item {
  margin-bottom: 18px;
}
```

```
.report-label {
  display: flex;
  justify-content: space-between;
  margin-bottom: 7px;
  font-size: 13px;
}
```

```
/* =====
NOTIFICATION
===== */
```

```
.toast {
  position: fixed;
  right: 25px;
  bottom: 25px;
  background: var(--card);
  border: 1px solid var(--border);
  box-shadow: var(--shadow);
}
```

```
padding: 14px 18px;
border-radius: 10px;
display: none;
z-index: 999;
}
```

```
.toast.show {
  display: block;
}
```

```
/* =====
MOBILE
===== */
```

```
.mobile-menu {
  display: none;
}
```

```
@media(max-width: 1000px) {
  .cards {
    grid-template-columns: repeat(2, 1fr);
  }
}
```

```
.form-grid {
  grid-template-columns: repeat(2, 1fr);
}
}
```

```
@media(max-width: 700px) {
  .sidebar {
    transform: translateX(-100%);
  }
}
```

```
    transition: 0.25s;
}
```

```
.sidebar.open {
    transform: translateX(0);
}
```

```
.main {
    margin-left: 0;
    width: 100%;
}
```

```
.mobile-menu {
    display: block;
    border: none;
    background: transparent;
    font-size: 22px;
    color: var(--text);
}
```

```
.cards {
    grid-template-columns: 1fr;
}
```

```
.form-grid {
    grid-template-columns: 1fr;
}
```

```
.content {
    padding: 15px;
}
```

```

.topbar {
  padding: 0 15px;
}

}

/* =====
LOGIN THEME BUTTON
===== */

.auth-theme {
  position: fixed;
  top: 20px;
  right: 20px;
}

.hidden {
  display: none !important;
}

/* =====
ENHANCED VISUALS, ANIMATIONS & COLORFUL SUBTOPICS
===== */

:root {
  --heading-color: #7c3aed;
  --heading-secondary: #0891b2;
  --accent-pink: #db2777;
  --accent-orange: #ea580c;
  --accent-green: #059669;

```

```
}
```

```
[data-theme="dark"] {  
  --heading-color: #a78bfa;  
  --heading-secondary: #22d3ee;  
  --accent-pink: #f472b6;  
  --accent-orange: #fb923c;  
  --accent-green: #34d399;  
}
```

```
@keyframes fadeUpERP {  
  from { opacity: 0; transform: translateY(18px); }  
  to { opacity: 1; transform: translateY(0); }  
}
```

```
@keyframes slideInERP {  
  from { opacity: 0; transform: translateX(-18px); }  
  to { opacity: 1; transform: translateX(0); }  
}
```

```
@keyframes floatERP {  
  0%, 100% { transform: translateY(0); }  
  50% { transform: translateY(-5px); }  
}
```

```
@keyframes pulseERP {  
  0%, 100% { box-shadow: 0 0 0 0 rgba(37,99,235,.15); }  
  50% { box-shadow: 0 0 0 10px rgba(37,99,235,0); }  
}
```

```
@keyframes shimmerERP {
```

```
0% { background-position: -300px 0; }
100% { background-position: 300px 0; }
}

.auth-box {
  animation: fadeUpERP .65s ease both;
}

.logo {
  animation: floatERP 3s ease-in-out infinite, pulseERP 2.5s infinite;
}

.sidebar .brand {
  background: linear-gradient(110deg, transparent 25%, rgba(255,255,255,.08) 50%, transparent 75%);
  background-size: 300px 100%;
  animation: shimmerERP 5s linear infinite;
}

.page:not(.hidden) {
  animation: fadeUpERP .4s ease both;
}

.stat-card {
  transition: transform .25s ease, box-shadow .25s ease, border-color .25s ease;
}

.stat-card:hover {
  transform: translateY(-6px);
  box-shadow: 0 12px 30px rgba(37,99,235,.14);
  border-color: var(--primary);
}
```

```
}
```

```
.stat-card:nth-child(1) .icon { color: var(--primary); }
```

```
.stat-card:nth-child(2) .icon { color: var(--accent-orange); }
```

```
.stat-card:nth-child(3) .icon { color: var(--accent-green); }
```

```
.stat-card:nth-child(4) .icon { color: var(--accent-pink); }
```

```
.section {
```

```
  transition: transform .25s ease, box-shadow .25s ease;
```

```
}
```

```
.section:hover {
```

```
  box-shadow: 0 10px 28px rgba(0,0,0,.10);
```

```
}
```

```
.section-header h2 {
```

```
  background: linear-gradient(90deg, var(--heading-color), var(--heading-secondary), var(--accent-pink));
```

```
  background-size: 200% auto;
```

```
  -webkit-background-clip: text;
```

```
  background-clip: text;
```

```
  -webkit-text-fill-color: transparent;
```

```
  animation: headingGradientERP 6s linear infinite;
```

```
}
```

```
@keyframes headingGradientERP {
```

```
  to { background-position: 200% center; }
```

```
}
```

```
.nav-section:nth-of-type(1) { color: #60a5fa; }
```

```
.nav-section:nth-of-type(2) { color: #34d399; }
```

```
.nav-section:nth-of-type(3) { color: #fbbf24; }
```

```
.btn {  
    position: relative;  
    overflow: hidden;  
    transition: transform .2s ease, box-shadow .2s ease, filter .2s ease;  
}
```

```
.btn:hover {  
    transform: translateY(-2px);  
    filter: brightness(1.05);  
    box-shadow: 0 7px 18px rgba(37,99,235,.18);  
}
```

```
.btn:active {  
    transform: scale(.97);  
}
```

```
.nav button {  
    transition: transform .2s ease, background .2s ease, color .2s ease;  
}
```

```
.nav button:hover {  
    transform: translateX(4px);  
}
```

```
tbody tr {  
    transition: background .2s ease, transform .2s ease;  
}
```

```
tbody tr:hover {
```

```
    transform: translateX(2px);  
  }
```

```
.info-grid {  
  display: grid;  
  grid-template-columns: repeat(3, 1fr);  
  gap: 18px;  
  margin-bottom: 20px;  
}
```

```
.info-card {  
  padding: 20px;  
  border-radius: 14px;  
  background: linear-gradient(135deg, var(--card), rgba(37,99,235,.06));  
  border: 1px solid var(--border);  
  transition: transform .25s ease, box-shadow .25s ease;  
}
```

```
.info-card:hover {  
  transform: translateY(-5px);  
  box-shadow: var(--shadow);  
}
```

```
.info-card h3 {  
  margin-bottom: 8px;  
  color: var(--heading-color);  
}
```

```
.info-card p {  
  color: var(--muted);  
  line-height: 1.6;
```

```
    font-size: 13px;
}
```

```
.feature-list {
    display: grid;
    grid-template-columns: repeat(2, 1fr);
    gap: 12px;
}
```

```
.feature-item {
    padding: 13px;
    border: 1px solid var(--border);
    border-radius: 10px;
    color: var(--text);
    background: rgba(148,163,184,.04);
}
```

```
.feature-item strong { color: var(--heading-secondary); }
```

```
.forgot-link {
    display: block;
    text-align: right;
    margin: 9px 0 2px;
    border: none;
    background: transparent;
    color: var(--accent-pink);
    font-size: 13px;
    font-weight: 700;
}
```

```
.forgot-link:hover { text-decoration: underline; }
```

```
.security-note {  
    margin: 12px 0 18px;  
    padding: 11px;  
    border-left: 4px solid var(--warning);  
    background: rgba(217,119,6,.09);  
    color: var(--muted);  
    font-size: 12px;  
    line-height: 1.5;  
    border-radius: 6px;  
}
```

```
.about-hero {  
    padding: 28px;  
    border-radius: 16px;  
    margin-bottom: 20px;  
    color: white;  
    background: linear-gradient(135deg, #2563eb, #7c3aed, #db2777);  
    background-size: 200% 200%;  
    animation: heroGradientERP 8s ease infinite;  
}
```

```
@keyframes heroGradientERP {  
    0%,100% { background-position: 0% 50%; }  
    50% { background-position: 100% 50%; }  
}
```

```
.about-hero h2 {  
    font-size: 27px;  
    margin-bottom: 8px;  
}
```

```
.about-hero p { line-height: 1.7; opacity: .94; }
```

```
.timeline {  
  border-left: 3px solid var(--primary);  
  padding-left: 20px;  
}
```

```
.timeline-item {  
  position: relative;  
  margin-bottom: 18px;  
  animation: slideInERP .5s ease both;  
}
```

```
.timeline-item::before {  
  content: "";  
  position: absolute;  
  left: -29px;  
  top: 4px;  
  width: 12px;  
  height: 12px;  
  border-radius: 50%;  
  background: var(--accent-pink);  
  border: 3px solid var(--card);  
  box-shadow: 0 0 0 2px var(--accent-pink);  
}
```

```
.timeline-item strong { color: var(--heading-color); }
```

```
@media(max-width: 1000px) {  
  .info-grid { grid-template-columns: 1fr 1fr; }
```

```
}
```

```
@media(max-width: 700px) {  
  .info-grid, .feature-list { grid-template-columns: 1fr; }  
  .about-hero h2 { font-size: 22px; }  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<!-- =====  
  LOGIN  
  ===== -->
```

```
<div id="loginScreen" class="auth-container">
```

```
<button class="theme-toggle auth-theme" onclick="toggleTheme()" id="authThemeBtn">
```

```
  🌙 Dark
```

```
</button>
```

```
<div class="auth-box">
```

```
<div class="logo">ERP</div>
```

```
<h1>ERP System</h1>
```

```
<p class="auth-subtitle">
```

```
  Enterprise Resource Planning
```

```
</p>
```

```
<div id="loginForm">
```

```
  <div class="form-group">
```

```
    <label>Username</label>
```

```
    <input id="loginUsername"
```

```
      type="text"
```

```
      placeholder="Enter username">
```

```
  </div>
```

```
  <div class="form-group">
```

```
    <label>Password</label>
```

```
    <input id="loginPassword"
```

```
      type="password"
```

```
      placeholder="Enter password">
```

```
  </div>
```

```
  <button class="forgot-link" onclick="showForgotPassword()">
```

```
    Forgot Password?
```

```
  </button>
```

```
  <button class="btn btn-primary btn-full"
```

```
    onclick="login()">
```

```
    Login
```

```
  </button>
```

```
<div class="demo-login">
```

```
  <strong>Demo Admin</strong><br>
```

```
  Username: admin<br>
```

```
  Password: Admin@123
```

```
</div>
```

```
<div class="auth-switch">  
  Don't have an account?  
  <button onclick="showRegister()">  
    Create an Account  
  </button>  
</div>
```

```
</div>
```

```
<!-- FORGOT PASSWORD -->
```

```
<div id="forgotPasswordForm" class="hidden">
```

```
  <h2 style="margin-bottom:10px;">Reset Password</h2>
```

```
  <p class="auth-subtitle" style="margin-top:0;">Reset your demo account password</p>
```

```
  <div class="security-note">
```

```
    Demo mode: this reset is performed locally in your browser.
```

```
    A real production ERP should use verified email/OTP recovery.
```

```
  </div>
```

```
  <div class="form-group">
```

```
    <label>Username</label>
```

```
    <input id="resetUsername" type="text" placeholder="Enter your username">
```

```
  </div>
```

```
  <div class="form-group">
```

```
    <label>Registered Email</label>
```

```
    <input id="resetEmail" type="email" placeholder="Enter your registered email">
```

```
  </div>
```

```
<div class="form-group">
  <label>New Password</label>
  <input id="resetPassword" type="password" placeholder="Create new password">
</div>
```

```
<div class="form-group">
  <label>Confirm New Password</label>
  <input id="resetConfirm" type="password" placeholder="Confirm new password">
</div>
```

```
<button class="btn btn-warning btn-full" onclick="resetPassword()">
  Reset Password
</button>
```

```
<div class="auth-switch">
  Remembered your password?
  <button onclick="showLogin()">Back to Login</button>
</div>
```

```
<div class="auth-switch" style="margin-top:10px;">
  Don't have an account?
  <button onclick="showRegister()">Create Account</button>
</div>
```

```
</div>
```

```
<!-- REGISTER -->
```

```
<div id="registerForm" class="hidden">
```

```
<h2 style="margin-bottom:18px;">Create Account</h2>
```

```
<div class="form-group">
  <label>Full Name</label>
  <input id="regName"
    type="text"
    placeholder="Enter full name">
</div>

<div class="form-group">
  <label>Email</label>
  <input id="regEmail"
    type="email"
    placeholder="Enter email">
</div>

<div class="form-group">
  <label>Username</label>
  <input id="regUsername"
    type="text"
    placeholder="Create username">
</div>

<div class="form-group">
  <label>Password</label>
  <input id="regPassword"
    type="password"
    placeholder="Create password">

  <div class="password-note">
    Minimum 6 characters.
  </div>
```

```
</div>
```

```
<div class="form-group">
```

```
<label>Confirm Password</label>
```

```
<input id="regConfirm"
```

```
  type="password"
```

```
  placeholder="Confirm password">
```

```
</div>
```

```
<button class="btn btn-success btn-full"
```

```
  onclick="registerUser()">
```

```
  Create Account
```

```
</button>
```

```
<div class="auth-switch">
```

```
  Already have an account?
```

```
<button onclick="showLogin()">
```

```
  Back to Login
```

```
</button>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<!-- =====
```

```
MAIN APPLICATION
```

```
===== -->
```

```
<div id="app">
```

```
<div class="layout">
```

```
<!-- SIDEBAR -->
```

```
<aside class="sidebar" id="sidebar">
```

```
<div class="brand">
```

```
<span>ERP</span> System
```

```
</div>
```

```
<div class="user-info">
```

```
<div class="user-avatar" id="userAvatar">
```

```
A
```

```
</div>
```

```
<div class="user-name" id="sidebarUser">
```

```
Admin
```

```
</div>
```

```
<div class="user-role">
```

```
Administrator
```

```
</div>
```

```
<div style="clear:both;"></div>
```

```
</div>
```

```
<div class="nav">
```

```
<div class="nav-section">
```

```
  Main
```

```
</div>
```

```
<button class="active"
```

```
  onclick="showPage('dashboard',this)">
```

```
     Dashboard
```

```
</button>
```

```
<button onclick="showPage('employees',this)">
```

```
   HR Management
```

```
</button>
```

```
<button onclick="showPage('inventory',this)">
```

```
   Inventory
```

```
</button>
```

```
<button onclick="showPage('customers',this)">
```

```
   Customers
```

```
</button>
```

```
<button onclick="showPage('sales',this)">
```

```
   Sales
```

```
</button>
```

```
<div class="nav-section">
```

```
  Finance
```

```
</div>
```

```
<button onclick="showPage('accounting',this)">
```

 Accounting

</button>

<button onclick="showPage('procurement',this)">

 Procurement

</button>

<div class="nav-section">

Administration

</div>

<button onclick="showPage('reports',this)">

 Reports

</button>

<button onclick="showPage('users',this)">

 Users

</button>

<button onclick="showPage('audit',this)">

 Audit Logs

</button>

<button onclick="showPage('about',this)">

 About ERP

</button>

</div>

<div class="sidebar-bottom">

```
<button class="btn btn-secondary btn-full"
```

```
  onclick="logout()">
```

```
     Logout
```

```
</button>
```

```
</div>
```

```
</aside>
```

```
<!-- MAIN -->
```

```
<main class="main">
```

```
<header class="topbar">
```

```
<div style="display:flex;align-items:center;gap:12px;">
```

```
<button class="mobile-menu"
```

```
  onclick="toggleSidebar()">
```

```
    ≡
```

```
</button>
```

```
<div class="page-title" id="pageTitle">
```

```
  Dashboard
```

```
</div>
```

```
</div>
```

```
<div class="top-actions">
```

```
<button class="theme-toggle"
  onclick="toggleTheme()"
  id="themeBtn">
```

```
  🌙 Dark
```

```
</button>
```

```
<span id="topUser"></span>
```

```
</div>
```

```
</header>
```

```
<div class="content">
```

```
<!-- =====
DASHBOARD
===== -->
```

```
<section id="dashboardPage" class="page">
```

```
<div class="cards">
```

```
<div class="stat-card">
```

```
<div class="icon"> 👤 </div>
```

```
<div class="stat-title">
```

```
  Employees
```

```
</div>
```

```
<div class="stat-value"
```

```
  id="employeeCount">
```

```
0
</div>
</div>
```

```
<div class="stat-card">
  <div class="icon"> 📦 </div>
  <div class="stat-title">
    Products
  </div>
  <div class="stat-value"
    id="productCount">
    0
  </div>
</div>
```

```
<div class="stat-card">
  <div class="icon"> 👤 </div>
  <div class="stat-title">
    Customers
  </div>
  <div class="stat-value"
    id="customerCount">
    0
  </div>
</div>
```

```
<div class="stat-card">
  <div class="icon"> 💰 </div>
  <div class="stat-title">
    Total Sales
  </div>
```

```
<div class="stat-value"
  id="salesTotal">
  ₹0
</div>
</div>
```

```
</div>
```

```
<div class="section">
```

```
<div class="section-header">
  <h2>ERP Overview</h2>
  <button class="btn btn-primary"
    onclick="refreshDashboard()">
    Refresh
  </button>
</div>
```

```
<p style="color:var(--muted);">
  Welcome to the Enterprise Resource Planning
  dashboard. Manage your organization from one
  centralized system.
</p>
```

```
</div>
```

```
<div class="info-grid">
```

```
<div class="info-card">
```

⚡ Smart Operations</h3>

<p>Connect HR, inventory, sales, procurement and finance workflows in one centralized ERP workspace.</p>

</div>

<div class="info-card">

🔒 Security & Audit</h3>

<p>Track important actions with audit logs and keep your demo environment organized with user accounts.</p>

</div>

<div class="info-card">

📊 Business Insights</h3>

<p>Use dashboards, KPIs, sales totals and accounting summaries to understand operational performance.</p>

</div>

</div>

<div class="section">

<div class="section-header">

<h2>ERP Features</h2>

</div>

<div class="feature-list">

<div class="feature-item"> 👤 HR Management — Employee records and salary information.</div>

<div class="feature-item"> 📦 Inventory Control — Products, stock levels and availability.</div>

<div class="feature-item"> 💰 Sales Management — Customers, invoices and sales transactions.</div>

`<div class="feature-item">  <strong>Accounting</strong> — Income, expenses and balance tracking.</div>`

`<div class="feature-item">  <strong>Procurement</strong> — Supplier and purchase order tracking.</div>`

`<div class="feature-item">  <strong>Reporting</strong> — KPIs, analytics and business summaries.</div>`

`</div>`

`</div>`

`<div class="section">`

`<div class="section-header">`

`<h2>Recent Sales</h2>`

`</div>`

`<div class="table-container">`

`<table>`

`<thead>`

`<tr>`

`<th>Invoice</th>`

`<th>Customer</th>`

`<th>Amount</th>`

`<th>Date</th>`

`</tr>`

`</thead>`

`<tbody id="recentSalesTable">`

`</tbody>`

</table>

</div>

</div>

</section>

<!-- =====
HR
===== -->

<section id="employeesPage"
class="page hidden">

<div class="section">

<div class="section-header">
<h2>Employee Management</h2>
</div>

<div class="form-grid">

<div class="form-group">
<label>Name</label>
<input id="empName"
placeholder="Employee name">
</div>

<div class="form-group">

```
<label>Department</label>
<input id="empDept"
      placeholder="Department">
</div>
```

```
<div class="form-group">
  <label>Position</label>
  <input id="empPosition"
        placeholder="Position">
</div>
```

```
<div class="form-group">
  <label>Email</label>
  <input id="empEmail"
        placeholder="Email">
</div>
```

```
<div class="form-group">
  <label>Salary</label>
  <input id="empSalary"
        type="number"
        placeholder="Salary">
</div>
```

```
</div>
```

```
<div class="form-actions">
  <button class="btn btn-primary"
        onclick="addEmployee()">
    Add Employee
  </button>
```

```
</div>
```

```
</div>
```

```
<div class="section">
```

```
<div class="section-header">
```

```
<h2>Employees</h2>
```

```
<input style="max-width:250px;"
```

```
placeholder="Search employees..."
```

```
onkeyup="searchTable(this,'employeesTable')">
```

```
</div>
```

```
<div class="table-container">
```

```
<table id="employeesTable">
```

```
<thead>
```

```
<tr>
```

```
<th>ID</th>
```

```
<th>Name</th>
```

```
<th>Department</th>
```

```
<th>Position</th>
```

```
<th>Email</th>
```

```
<th>Salary</th>
```

```
<th>Action</th>
```

```
</tr>
```

```
</thead>
```

```
<tbody id="employeeTableBody">
```

```
</tbody>
```

```
</table>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<!-- =====
```

```
INVENTORY
```

```
===== -->
```

```
<section id="inventoryPage"
```

```
class="page hidden">
```

```
<div class="section">
```

```
<div class="section-header">
```

```
<h2>Add Product</h2>
```

```
</div>
```

```
<div class="form-grid">
```

```
<div class="form-group">
```

```
<label>Product Name</label>
```

```
<input id="productName"
```

```
placeholder="Product name">
```

```
</div>
```

```
<div class="form-group">  
  <label>Category</label>  
  <input id="productCategory"  
    placeholder="Category">  
</div>
```

```
<div class="form-group">  
  <label>Price</label>  
  <input id="productPrice"  
    type="number"  
    placeholder="Price">  
</div>
```

```
<div class="form-group">  
  <label>Stock</label>  
  <input id="productStock"  
    type="number"  
    placeholder="Stock">  
</div>
```

```
</div>
```

```
<div class="form-actions">
```

```
  <button class="btn btn-primary"  
    onclick="addProduct()">  
    Add Product  
  </button>
```

</div>

</div>

<div class="section">

<div class="section-header">

<h2>Inventory</h2>

</div>

<div class="table-container">

<table>

<thead>

<tr>

<th>ID</th>

<th>Product</th>

<th>Category</th>

<th>Price</th>

<th>Stock</th>

<th>Status</th>

<th>Action</th>

</tr>

</thead>

<tbody id="productTableBody">

</tbody>

</table>

</div>

</div>

</section>

```
<!-- =====  
CUSTOMERS  
===== -->
```

```
<section id="customersPage"  
  class="page hidden">
```

```
<div class="section">
```

```
<div class="section-header">  
  <h2>Add Customer</h2>  
</div>
```

```
<div class="form-grid">
```

```
<div class="form-group">  
  <label>Name</label>  
  <input id="customerName"  
    placeholder="Customer name">  
</div>
```

```
<div class="form-group">  
  <label>Email</label>
```

```
        <input id="customerEmail"
              placeholder="Email">
    </div>
```

```
<div class="form-group">
    <label>Phone</label>
    <input id="customerPhone"
          placeholder="Phone">
</div>
```

```
<div class="form-group">
    <label>Company</label>
    <input id="customerCompany"
          placeholder="Company">
</div>
```

```
</div>
```

```
<div class="form-actions">
```

```
    <button class="btn btn-primary"
            onclick="addCustomer()">
        Add Customer
    </button>
```

```
</div>
```

```
</div>
```

```
<div class="section">
```

```
<div class="section-header">
```

```
  <h2>Customers</h2>
```

```
</div>
```

```
<div class="table-container">
```

```
  <table>
```

```
    <thead>
```

```
      <tr>
```

```
        <th>ID</th>
```

```
        <th>Name</th>
```

```
        <th>Email</th>
```

```
        <th>Phone</th>
```

```
        <th>Company</th>
```

```
        <th>Action</th>
```

```
      </tr>
```

```
    </thead>
```

```
    <tbody id="customerTableBody">
```

```
  </tbody>
```

```
</table>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<!-- =====  
SALES  
===== -->
```

```
<section id="salesPage"  
  class="page hidden">
```

```
<div class="section">
```

```
<div class="section-header">
```

```
<h2>Create Sale</h2>
```

```
</div>
```

```
<div class="form-grid">
```

```
<div class="form-group">
```

```
<label>Customer (Manual)</label>
```

```
<input id="saleCustomerManual"
```

```
  type="text"
```

```
  placeholder="Type customer name">
```

```
</div>
```

```
<div class="form-group">
```

```
<label>Product (Manual)</label>
```

```
<input id="saleProductManual"
```

```
  type="text"
```

```
  placeholder="Type product name">
```

```
</div>
```

```
<div class="form-group">
```

```
<label>Quantity</label>
<input id="saleQuantity"
      type="number"
      value="1"
      min="1"
      oninput="updateManualSaleTotal()">
</div>
```

```
<div class="form-group">
  <label>Unit Price (₹)</label>
  <input id="saleUnitPrice"
        type="number"
        min="0"
        step="0.01"
        placeholder="Type price"
        oninput="updateManualSaleTotal()">
</div>
```

```
<div class="form-group">
  <label>Total (₹)</label>
  <input id="saleTotal"
        readonly
        value="0">
</div>
```

```
</div>
```

```
<div class="form-actions">
```

```
  <button class="btn btn-success"
          onclick="createSale()">
```

Create Sale

</button>

</div>

</div>

<div class="section">

<div class="section-header">

<h2>Sales History</h2>

</div>

<div class="table-container">

<table>

<thead>

<tr>

<th>Invoice</th>

<th>Customer</th>

<th>Product</th>

<th>Quantity</th>

<th>Total</th>

<th>Date</th>

<th>Action</th>

</tr>

</thead>

<tbody id="salesTableBody">

</tbody>

</table>

</div>

</div>

</section>

<!-- =====
ACCOUNTING
===== -->

<section id="accountingPage"
class="page hidden">

<div class="section">

<div class="section-header">
<h2>Add Transaction</h2>
</div>

<div class="form-grid">

<div class="form-group">
<label>Description</label>
<input id="transactionDescription"
placeholder="Description">
</div>

```
<div class="form-group">
  <label>Type</label>
  <select id="transactionType">
    <option value="Income">Income</option>
    <option value="Expense">Expense</option>
  </select>
</div>
```

```
<div class="form-group">
  <label>Amount</label>
  <input id="transactionAmount"
    type="number"
    placeholder="Amount">
</div>
```

```
</div>
```

```
<div class="form-actions">

  <button class="btn btn-primary"
    onclick="addTransaction()">
    Add Transaction
  </button>

</div>
```

```
</div>
```

```
<div class="section">
```

```
<div class="section-header">
```

```
  <h2>Transactions</h2>
```

```
</div>
```

```
<div class="table-container">
```

```
  <table>
```

```
    <thead>
```

```
      <tr>
```

```
        <th>ID</th>
```

```
        <th>Description</th>
```

```
        <th>Type</th>
```

```
        <th>Amount</th>
```

```
        <th>Date</th>
```

```
        <th>Action</th>
```

```
      </tr>
```

```
    </thead>
```

```
    <tbody id="transactionTableBody">
```

```
  </tbody>
```

```
</table>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<!-- =====  
PROCUREMENT  
===== -->
```

```
<section id="procurementPage"  
  class="page hidden">
```

```
<div class="section">
```

```
<div class="section-header">  
  <h2>Create Purchase Order</h2>  
</div>
```

```
<div class="form-grid">
```

```
<div class="form-group">  
  <label>Supplier</label>  
  <input id="purchaseSupplier"  
    placeholder="Supplier">  
</div>
```

```
<div class="form-group">  
  <label>Product</label>  
  <input id="purchaseProduct"  
    placeholder="Product">  
</div>
```

```
<div class="form-group">  
  <label>Quantity</label>  
  <input id="purchaseQuantity"
```

```
        type="number"
        placeholder="Quantity">
    </div>
```

```
<div class="form-group">
    <label>Amount</label>
    <input id="purchaseAmount"
        type="number"
        placeholder="Amount">
</div>
```

```
</div>
```

```
<div class="form-actions">
```

```
    <button class="btn btn-primary"
        onclick="addPurchase()">
        Create Purchase
    </button>
```

```
</div>
```

```
</div>
```

```
<div class="section">
```

```
    <div class="section-header">
        <h2>Purchase Orders</h2>
    </div>
```

```
<div class="table-container">
```

```
<table>
```

```
<thead>
```

```
<tr>
```

```
<th>PO Number</th>
```

```
<th>Supplier</th>
```

```
<th>Product</th>
```

```
<th>Quantity</th>
```

```
<th>Amount</th>
```

```
<th>Date</th>
```

```
<th>Action</th>
```

```
</tr>
```

```
</thead>
```

```
<tbody id="purchaseTableBody">
```

```
</tbody>
```

```
</table>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<!-- =====
```

```
REPORTS
```

```
===== -->
```

```
<section id="reportsPage"
  class="page hidden">
```

```
<div class="section">
```

```
<div class="section-header">
```

```
<div>
```

```
<h2>Export Business Data</h2>
```

```
<p style="color:var(--muted);margin-top:6px;">Download the current ERP business
data as a PDF report.</p>
```

```
</div>
```

```
<div class="form-actions" style="margin-top:0;">
```

```
<button class="btn btn-primary" onclick="downloadReportsPDF()">📄 Export Data
as PDF</button>
```

```
<button class="btn btn-secondary" onclick="window.print()">🖨️ Print
Report</button>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<div class="cards">
```

```
<div class="stat-card">
```

```
<div class="stat-title">
```

```
  Total Income
```

```
</div>
```

```
<div class="stat-value"
```

```
  id="reportIncome">
```

```
    ₹0
```

```
</div>
```

```
</div>
```

```
<div class="stat-card">
  <div class="stat-title">
    Total Expenses
  </div>

  <div class="stat-value"
    id="reportExpense">
    ₹0
  </div>
</div>
```

```
<div class="stat-card">
  <div class="stat-title">
    Net Balance
  </div>

  <div class="stat-value"
    id="reportBalance">
    ₹0
  </div>
</div>
```

```
<div class="stat-card">
  <div class="stat-title">
    Total Orders
  </div>

  <div class="stat-value"
    id="reportOrders">
    0
  </div>
</div>
```

</div>

</div>

</div>

<div class="section">

<div class="section-header">

<h2>Business Analytics</h2>

</div>

<div class="report-item">

<div class="report-label">

Sales Performance

0%

</div>

<div class="progress">

<div class="progress-bar"

id="salesProgress"

style="width:0%;">

</div>

</div>

</div>

<div class="report-item">

```
<div class="report-label">
  <span>Inventory Utilization</span>
  <span id="inventoryPercent">0%</span>
</div>
```

```
<div class="progress">
  <div class="progress-bar"
    id="inventoryProgress"
    style="width:0%;">
  </div>
</div>
```

```
</div>
```

```
<div class="report-item">
```

```
<div class="report-label">
  <span>Customer Growth</span>
  <span id="customerPercent">0%</span>
</div>
```

```
<div class="progress">
  <div class="progress-bar"
    id="customerProgress"
    style="width:0%;">
  </div>
</div>
```

```
</div>
```

</div>

</section>

```
<!-- =====  
USERS  
===== -->
```

```
<section id="usersPage"  
  class="page hidden">
```

```
<div class="section">
```

```
<div class="section-header">  
  <h2>System Users</h2>  
</div>
```

```
<div class="table-container">
```

```
<table>
```

```
<thead>
```

```
<tr>
```

```
<th>Name</th>
```

```
<th>Email</th>
```

```
<th>Username</th>
```

```
<th>Role</th>
```

```
</tr>
```

```
</thead>
```

```
<tbody id="usersTableBody">
```

```
</tbody>
```

```
</table>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<!-- =====
```

```
AUDIT
```

```
===== -->
```

```
<section id="auditPage"
```

```
class="page hidden">
```

```
<div class="section">
```

```
<div class="section-header">
```

```
<h2>Audit Logs</h2>
```

```
<button class="btn btn-danger"
```

```
onclick="clearAudit()">
```

```
Clear Logs
```

```
</button>
```

```
</div>
```

```
<div class="table-container">
```

```
<table>
```

```
<thead>
```

```
<tr>
```

```
<th>Date</th>
```

```
<th>User</th>
```

```
<th>Action</th>
```

```
<th>Details</th>
```

```
</tr>
```

```
</thead>
```

```
<tbody id="auditTableBody">
```

```
</tbody>
```

```
</table>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<!-- =====
```

```
ABOUT ERP
```

```
===== -->
```

```
<section id="aboutPage" class="page hidden">
```

```
<div class="about-hero">
```

```
  <h2>Enterprise Resource Planning System</h2>
```

```
  <p>A single-file ERP project designed to demonstrate how business departments can  
work together through a centralized digital platform.</p>
```

```
</div>
```

```
<div class="info-grid">
```

```
  <div class="info-card"><h3>🎯 Purpose</h3><p>Centralize operational information and  
provide a simple interface for learning ERP concepts.</p></div>
```

```
  <div class="info-card"><h3>🏢 Departments</h3><p>HR, Inventory, Customers, Sales,  
Accounting, Procurement, Reporting and Administration.</p></div>
```

```
  <div class="info-card"><h3>💻 Technology</h3><p>HTML, CSS and JavaScript with  
browser localStorage/sessionStorage for this demonstration.</p></div>
```

```
</div>
```

```
<div class="section">
```

```
  <div class="section-header"><h2>System Information</h2></div>
```

```
  <div class="feature-list">
```

```
    <div class="feature-item">🌙 <strong>Theme System</strong> — Dark and light  
modes with saved preference.</div>
```

```
    <div class="feature-item">🔑 <strong>Account System</strong> — Login, registration  
and demo password recovery.</div>
```

```
    <div class="feature-item">📱 <strong>Responsive UI</strong> — Designed for  
desktop, tablet and mobile screens.</div>
```

```
    <div class="feature-item">✨ <strong>Animations</strong> — Page transitions, hover  
effects, gradients and motion.</div>
```

```
    <div class="feature-item">📁 <strong>Local Data</strong> — Demo records persist in  
the browser.</div>
```

```
    <div class="feature-item">🚀 <strong>Project Ready</strong> — Suitable as a front-  
end academic ERP demonstration.</div>
```

```
  </div>
```

```
</div>
```

```
<div class="section">
```

```
<div class="section-header"><h2>ERP Workflow</h2></div>
```

```
<div class="timeline">
```

```
    <div class="timeline-item"><strong>1. Create users</strong><br><span
style="color:var(--muted);">Register an account or use the demo administrator.</span></div>
```

```
    <div class="timeline-item"><strong>2. Configure operations</strong><br><span
style="color:var(--muted);">Add employees, products and customers.</span></div>
```

```
    <div class="timeline-item"><strong>3. Process business</strong><br><span
style="color:var(--muted);">Create sales, purchases and accounting transactions.</span></div>
```

```
    <div class="timeline-item"><strong>4. Analyze results</strong><br><span
style="color:var(--muted);">Review dashboard KPIs, reports and audit activity.</span></div>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
</div>
```

```
</main>
```

```
</div>
```

```
</div>
```

```
<div id="toast" class="toast"></div>
```

```
<script>
```

```
/* =====
```

```
    DATABASE
```

```
    ===== */
```

```
const DB_KEY = "ERP_DATABASE";
const USERS_KEY = "ERP_USERS";
const SESSION_KEY = "ERP_LOGGED_USER";
const THEME_KEY = "ERP_THEME";

let db;

/* =====
DEFAULT DATABASE
===== */

function initializeDatabase() {

    const existing = localStorage.getItem(DB_KEY);

    if (existing) {

        db = JSON.parse(existing);

        if (!db.employees) db.employees = [];
        if (!db.products) db.products = [];
        if (!db.customers) db.customers = [];
        if (!db.sales) db.sales = [];
        if (!db.transactions) db.transactions = [];
        if (!db.purchases) db.purchases = [];
        if (!db.audit) db.audit = [];

        saveDB();

        return;
    }
}
```

```
}
```

```
db = {
```

```
  employees: [
```

```
    {
```

```
      id: 1,
```

```
      name: "Admin Employee",
```

```
      department: "Management",
```

```
      position: "Administrator",
```

```
      email: "admin@erp.com",
```

```
      salary: 75000
```

```
    }
```

```
  ],
```

```
  products: [
```

```
    {
```

```
      id: 1,
```

```
      name: "Laptop",
```

```
      category: "Electronics",
```

```
      price: 65000,
```

```
      stock: 20
```

```
    },
```

```
    {
```

```
      id: 2,
```

```
      name: "Monitor",
```

```
      category: "Electronics",
```

```
      price: 15000,
```

```
      stock: 35
```

```
    },
```

```
    {
```

```
        id: 3,
        name: "Keyboard",
        category: "Accessories",
        price: 2500,
        stock: 50
    }
],

customers: [
    {
        id: 1,
        name: "ABC Technologies",
        email: "contact@abc.com",
        phone: "9876543210",
        company: "ABC Technologies"
    }
],

sales: [],

transactions: [],

purchases: [],

audit: []

};

saveDB();
}
```

```
/* =====  
USERS  
===== */  
  
function initializeUsers() {  
  
    let users = JSON.parse(  
        localStorage.getItem(USERS_KEY) || "[]"  
    );  
  
    if (!users.some(u => u.username === "admin")) {  
  
        users.push({  
            name: "Administrator",  
            email: "admin@erp.com",  
            username: "admin",  
            password: "Admin@123",  
            role: "Administrator"  
        });  
  
    }  
  
    localStorage.setItem(  
        USERS_KEY,  
        JSON.stringify(users)  
    );  
}  
  
/* =====
```

SAVE DATABASE

===== */

```
function saveDB() {
```

```
    localStorage.setItem(
```

```
        DB_KEY,
```

```
        JSON.stringify(db)
```

```
    );
```

```
}
```

```
/* =====
```

THEME

===== */

```
function initializeTheme() {
```

```
    const savedTheme =
```

```
        localStorage.getItem(THEME_KEY) || "light";
```

```
    document.documentElement.setAttribute(
```

```
        "data-theme",
```

```
        savedTheme
```

```
    );
```

```
    updateThemeButtons();
```

```
}
```

```
function toggleTheme() {

    const current =
        document.documentElement.getAttribute("data-theme")
        || "light";

    const next =
        current === "dark" ? "light" : "dark";

    document.documentElement.setAttribute(
        "data-theme",
        next
    );

    localStorage.setItem(
        THEME_KEY,
        next
    );

    updateThemeButtons();
}
```

```
function updateThemeButtons() {

    const theme =
        document.documentElement.getAttribute("data-theme")
        || "light";

    const buttons =
        document.querySelectorAll(".theme-toggle");
```

```

buttons.forEach(button => {

    if (theme === "dark") {
        button.innerHTML = "☀️ Light";
    } else {
        button.innerHTML = "🌙 Dark";
    }

});

}

/* =====
AUTH
===== */

function showRegister() {

    document.getElementById("loginForm")
        .classList.add("hidden");

    document.getElementById("forgotPasswordForm")
        .classList.add("hidden");

    document.getElementById("registerForm")
        .classList.remove("hidden");

}

```

```
function showLogin() {  
  
    document.getElementById("registerForm")  
        .classList.add("hidden");  
  
    document.getElementById("forgotPasswordForm")  
        .classList.add("hidden");  
  
    document.getElementById("loginForm")  
        .classList.remove("hidden");  
  
}
```

```
function showForgotPassword() {  
  
    document.getElementById("loginForm").classList.add("hidden");  
    document.getElementById("registerForm").classList.add("hidden");  
    document.getElementById("forgotPasswordForm").classList.remove("hidden");  
  
}
```

```
function resetPassword() {  
  
    const username = document.getElementById("resetUsername").value.trim();  
    const email = document.getElementById("resetEmail").value.trim();  
    const password = document.getElementById("resetPassword").value;  
    const confirm = document.getElementById("resetConfirm").value;
```

```
if (!username || !email || !password || !confirm) {
  showToast("Please complete all reset fields.");
  return;
}

if (password.length < 6) {
  showToast("Password must contain at least 6 characters.");
  return;
}

if (password !== confirm) {
  showToast("Passwords do not match.");
  return;
}

let users = JSON.parse(localStorage.getItem(USERS_KEY) || "[]");

const userIndex = users.findIndex(u =>
  u.username.toLowerCase() === username.toLowerCase() &&
  u.email.toLowerCase() === email.toLowerCase()
);

if (userIndex === -1) {
  showToast("Username and registered email do not match.");
  return;
}

users[userIndex].password = password;

localStorage.setItem(USERS_KEY, JSON.stringify(users));
```

```
addAudit(username, "Password Reset", "Password changed through demo recovery");

document.getElementById("resetUsername").value = "";
document.getElementById("resetEmail").value = "";
document.getElementById("resetPassword").value = "";
document.getElementById("resetConfirm").value = "";

showLogin();
showToast("Password reset successfully. Please login.");

}
```

```
function registerUser() {

    const name =
        document.getElementById("regName").value.trim();

    const email =
        document.getElementById("regEmail").value.trim();

    const username =
        document.getElementById("regUsername").value.trim();

    const password =
        document.getElementById("regPassword").value;

    const confirm =
        document.getElementById("regConfirm").value;
```

```
if (!name || !email || !username ||  
    !password || !confirm) {  
  
    showToast("Please fill all fields.");  
    return;  
}
```

```
if (password.length < 6) {  
  
    showToast(  
        "Password must contain at least 6 characters."  
    );  
  
    return;  
}
```

```
if (password !== confirm) {  
  
    showToast(  
        "Passwords do not match."  
    );  
  
    return;  
}
```

```
let users = JSON.parse(  
    localStorage.getItem(USERS_KEY) || "[]"
```

```
);
```

```
if (
```

```
  users.some(
```

```
    u => u.username.toLowerCase()
```

```
    === username.toLowerCase()
```

```
  )
```

```
){
```

```
  showToast("Username already exists.");
```

```
  return;
```

```
}
```

```
if (
```

```
  users.some(
```

```
    u => u.email.toLowerCase()
```

```
    === email.toLowerCase()
```

```
  )
```

```
){
```

```
  showToast("Email already exists.");
```

```
  return;
```

```
}
```

```
users.push({
```

```
        name,  
        email,  
        username,  
        password,  
        role: "User"  
    });
```

```
localStorage.setItem(  
    USERS_KEY,  
    JSON.stringify(users)  
);
```

```
addAudit(  
    username,  
    "Account Created",  
    "New user registered"  
);
```

```
document.getElementById("regName").value = "";  
document.getElementById("regEmail").value = "";  
document.getElementById("regUsername").value = "";  
document.getElementById("regPassword").value = "";  
document.getElementById("regConfirm").value = "";
```

```
showLogin();
```

```
    showToast(  
      "Account created successfully. Please login."  
    );  
  }  
}
```

```
function login() {  
  
  const username =  
    document.getElementById("loginUsername")  
    .value.trim();  
  
  const password =  
    document.getElementById("loginPassword")  
    .value;  
  
  const users = JSON.parse(  
    localStorage.getItem(USERS_KEY) || "[]"  
  );  
  
  const user = users.find(  
    u =>  
      u.username === username &&  
      u.password === password  
  );  
  
  if (!user) {
```

```
    showToast(  
        "Invalid username or password."  
    );  
  
    return;  
}
```

```
sessionStorage.setItem(  
    SESSION_KEY,  
    JSON.stringify(user)  
);
```

```
addAudit(  
    user.username,  
    "Login",  
    "User logged into ERP system"  
);
```

```
    openApplication(user);  
  
}
```

```
function openApplication(user) {  
  
    document.getElementById("loginScreen")  
        .style.display = "none";
```

```
document.getElementById("app")  
  .style.display = "block";
```

```
document.getElementById("sidebarUser")  
  .textContent = user.name;
```

```
document.getElementById("topUser")  
  .textContent = user.username;
```

```
document.getElementById("userAvatar")  
  .textContent =  
    user.name.charAt(0).toUpperCase();
```

```
refreshAll();
```

```
}
```

```
function logout() {
```

```
  const user =  
    JSON.parse(  
      sessionStorage.getItem(SESSION_KEY)  
    );
```

```
  if (user) {
```

```
    addAudit(
```

```

        user.username,
        "Logout",
        "User logged out"
    );

}

sessionStorage.removeItem(
    SESSION_KEY
);

document.getElementById("app")
    .style.display = "none";

document.getElementById("loginScreen")
    .style.display = "flex";

showLogin();

}

/* =====
NAVIGATION
===== */

function showPage(page, button) {

```

```
document.querySelectorAll(".page")
    .forEach(p => p.classList.add("hidden"));

const pageElement =
    document.getElementById(page + "Page");

if (pageElement) {
    pageElement.classList.remove("hidden");
}

document.querySelectorAll(".nav button")
    .forEach(b => b.classList.remove("active"));

if (button) {
    button.classList.add("active");
}

const titles = {

    dashboard: "Dashboard",
    employees: "HR Management",
    inventory: "Inventory",
    customers: "Customers",
    sales: "Sales",
    accounting: "Accounting",
    procurement: "Procurement",
    reports: "Reports",
```

```
users: "Users",  
audit: "Audit Logs",  
about: "About ERP"
```

```
};
```

```
document.getElementById("pageTitle")  
  .textContent = titles[page] || "ERP";
```

```
if (window.innerWidth <= 700) {
```

```
  document.getElementById("sidebar")  
    .classList.remove("open");
```

```
}
```

```
}
```

```
function toggleSidebar() {
```

```
  document.getElementById("sidebar")  
    .classList.toggle("open");
```

```
}
```

```
/* =====
```

```
EMPLOYEES
```

```
===== */
```

```
function addEmployee() {
```

```
    const name =
```

```
        document.getElementById("empName").value.trim();
```

```
    const department =
```

```
        document.getElementById("empDept").value.trim();
```

```
    const position =
```

```
        document.getElementById("empPosition").value.trim();
```

```
    const email =
```

```
        document.getElementById("empEmail").value.trim();
```

```
    const salary =
```

```
        Number(document.getElementById("empSalary").value);
```

```
    if (!name || !department || !position) {
```

```
        showToast(
```

```
            "Please enter employee details."
```

```
        );
```

```
        return;
```

```
    }
```

```
    db.employees.push({
```

```
        id: getNextId(db.employees),

        name,
        department,
        position,
        email,
        salary

    });

    saveDB();

    addAudit(
        getCurrentUser(),
        "Employee Added",
        name
    );

    document.getElementById("empName").value = "";
    document.getElementById("empDept").value = "";
    document.getElementById("empPosition").value = "";
    document.getElementById("empEmail").value = "";
    document.getElementById("empSalary").value = "";

    renderEmployees();
```

```
refreshDashboard();
```

```
showToast(  
  "Employee added successfully."  
);  
  
}
```

```
function renderEmployees() {
```

```
  const tbody =  
    document.getElementById("employeeTableBody");
```

```
  tbody.innerHTML = "";
```

```
  db.employees.forEach(emp => {
```

```
    tbody.innerHTML += `
```

```
    <tr>
```

```
      <td>${emp.id}</td>
```

```
      <td>${escapeHTML(emp.name)}</td>
```

```
      <td>${escapeHTML(emp.department)}</td>
```

```
      <td>${escapeHTML(emp.position)}</td>
```

```
<td>${escapeHTML(emp.email || "-")}</td>
```

```
<td>₹${formatNumber(emp.salary || 0)}</td>
```

```
<td><button class="btn btn-danger btn-small"
onclick="removeEmployee(${emp.id})">Remove</button></td>
```

```
</tr>
```

```
`;
```

```
});
```

```
}
```

```
/* =====
```

```
PRODUCTS
```

```
===== */
```

```
function addProduct() {
```

```
    const name =
```

```
        document.getElementById("productName")
```

```
        .value.trim();
```

```
    const category =
```

```
        document.getElementById("productCategory")
```

```
        .value.trim();
```

```
    const price =
```

```
Number(  
  document.getElementById("productPrice")  
  .value  
);
```

```
const stock =  
  Number(  
    document.getElementById("productStock")  
    .value  
  );
```

```
if (!name) {  
  
  showToast(  
    "Please enter product name."  
  );  
  
  return;  
}
```

```
db.products.push({  
  
  id: getNextId(db.products),  
  
  name,  
  category,  
  price,  
  stock
```

```
});
```

```
saveDB();
```

```
addAudit(  
    getCurrentUser(),  
    "Product Added",  
    name  
);
```

```
document.getElementById("productName").value = "";  
document.getElementById("productCategory").value = "";  
document.getElementById("productPrice").value = "";  
document.getElementById("productStock").value = "";
```

```
renderProducts();
```

```
refreshDashboard();
```

```
showToast(  
    "Product added successfully."  
);
```

```
}
```

```
function renderProducts() {
```

```
const tbody =
  document.getElementById("productTableBody");

tbody.innerHTML = "";

db.products.forEach(product => {

  let status;

  if (product.stock <= 0) {

    status =
      `<span class="badge badge-danger">Out of Stock</span>`;

  } else if (product.stock <= 10) {

    status =
      `<span class="badge badge-warning">Low Stock</span>`;

  } else {

    status =
      `<span class="badge badge-success">Available</span>`;

  }

  tbody.innerHTML += `
```

```

<tr>

    <td>${product.id}</td>

    <td>${escapeHTML(product.name)}</td>

    <td>${escapeHTML(product.category || "-")}</td>

    <td>₹${formatNumber(product.price)}</td>

    <td>${product.stock}</td>

    <td>${status}</td>

    <td><button class="btn btn-danger btn-small"
onclick="removeProduct(${product.id})">Remove</button></td>

</tr>

`;

});

updateSaleProductList();

}

/* =====
CUSTOMERS

```

```
===== */
```

```
function addCustomer() {

    const name =
        document.getElementById("customerName")
        .value.trim();

    const email =
        document.getElementById("customerEmail")
        .value.trim();

    const phone =
        document.getElementById("customerPhone")
        .value.trim();

    const company =
        document.getElementById("customerCompany")
        .value.trim();

    if (!name) {

        showToast(
            "Please enter customer name."
        );

        return;
    }
}
```

```
db.customers.push({
```

```
  id: getNextId(db.customers),
```

```
  name,
```

```
  email,
```

```
  phone,
```

```
  company
```

```
});
```

```
saveDB();
```

```
addAudit(
```

```
  getCurrentUser(),
```

```
  "Customer Added",
```

```
  name
```

```
);
```

```
document.getElementById("customerName").value = "";
```

```
document.getElementById("customerEmail").value = "";
```

```
document.getElementById("customerPhone").value = "";
```

```
document.getElementById("customerCompany").value = "";
```

```
renderCustomers();
```

```
refreshDashboard();
```

```
showToast(  
  "Customer added successfully."  
);  
  
}
```

```
function renderCustomers() {  
  
  const tbody =  
    document.getElementById("customerTableBody");  
  
  tbody.innerHTML = "";  
  
  db.customers.forEach(customer => {  
  
    tbody.innerHTML += `  
  
    <tr>  
  
      <td>${customer.id}</td>  
  
      <td>${escapeHTML(customer.name)}</td>  
  
      <td>${escapeHTML(customer.email || "-")}</td>  
  
      <td>${escapeHTML(customer.phone || "-")}</td>  
  
      <td>${escapeHTML(customer.company || "-")}</td>
```

```
        <td><button class="btn btn-danger btn-small"
onclick="removeCustomer(${customer.id})">Remove</button></td>
```

```
</tr>
```

```
`;
```

```
});
```

```
updateSaleCustomerList();
```

```
}
```

```
/* =====
```

```
SALES
```

```
===== */
```

```
function updateSaleCustomerList() {
```

```
    const select =
```

```
        document.getElementById("saleCustomer");
```

```
    if (!select) return;
```

```
    select.innerHTML =
```

```
        `<option value="">Select Customer</option>`;
```

```
db.customers.forEach(customer => {

  select.innerHTML += `

    <option value="${customer.id}">
      ${escapeHTML(customer.name)}
    </option>

  `;

});

}
```

```
function updateSaleProductList() {

  const select =
    document.getElementById("saleProduct");

  if (!select) return;

  select.innerHTML =
    `<option value="">Select Product</option>`;

  db.products.forEach(product => {

    select.innerHTML += `
```

```
<option value="{product.id}">
  {escapeHTML(product.name)}
  - ₹{formatNumber(product.price)}
  (Stock: {product.stock})
</option>
```

```
`;
```

```
});
```

```
}
```

```
function updateSalePrice() {
```

```
  const productId =
```

```
    Number(
      document.getElementById("saleProduct").value
    );
```

```
  const quantity =
```

```
    Number(
      document.getElementById("saleQuantity").value
    ) || 1;
```

```
  const product =
```

```
    db.products.find(
      p => p.id === productId
    );
```

```
const total =
  product
  ? product.price * quantity
  : 0;

document.getElementById("saleTotal")
  .value = total;

}

function updateManualSaleTotal() {
  const quantity = Number(document.getElementById("saleQuantity")?.value) || 0;
  const price = Number(document.getElementById("saleUnitPrice")?.value) || 0;
  const totalEl = document.getElementById("saleTotal");
  if (totalEl) totalEl.value = (quantity * price).toFixed(2);
}

function createSale() {
  const customerName = document.getElementById("saleCustomerManual").value.trim();
  const productName = document.getElementById("saleProductManual").value.trim();
  const quantity = Number(document.getElementById("saleQuantity").value);
  const unitPrice = Number(document.getElementById("saleUnitPrice").value);

  if (!customerName) {
    showToast("Please type the customer name.");
    return;
  }
}
```

```
if (!productName) {
  showToast("Please type the product name.");
  return;
}
if (!quantity || quantity <= 0) {
  showToast("Enter a valid quantity.");
  return;
}
if (unitPrice < 0 || !Number.isFinite(unitPrice)) {
  showToast("Enter a valid unit price.");
  return;
}

const matchingProduct = db.products.find(
  p => String(p.name).trim().toLowerCase() === productName.toLowerCase()
);

if (matchingProduct && matchingProduct.stock < quantity) {
  showToast("Insufficient stock for this inventory product.");
  return;
}

const total = unitPrice * quantity;
if (matchingProduct) matchingProduct.stock -= quantity;

const saleId = getNextId(db.sales);
const sale = {
  id: saleId,
  invoice: "INV-" + new Date().getFullYear() + "-" + String(saleId).padStart(4, "0"),
  customerId: null,
  customerName,
```

```
    productId: matchingProduct ? matchingProduct.id : null,  
    productName,  
    quantity,  
    unitPrice,  
    total,  
    date: new Date().toLocaleString()  
  };
```

```
db.sales.push(sale);  
db.transactions.push({  
  id: getNextId(db.transactions),  
  description: "Sale " + sale.invoice,  
  type: "Income",  
  amount: total,  
  date: new Date().toLocaleString()  
});
```

```
saveDB();  
addAudit(getCurrentUser(), "Sale Created", sale.invoice + " - " + customerName + " - " +  
productName);
```

```
renderSales();  
renderProducts();  
renderTransactions();  
refreshDashboard();  
refreshReports();
```

```
document.getElementById("saleCustomerManual").value = "";  
document.getElementById("saleProductManual").value = "";  
document.getElementById("saleQuantity").value = 1;  
document.getElementById("saleUnitPrice").value = "";
```

```
document.getElementById("saleTotal").value = "0";

showToast("Sale created successfully.");
}
```

```
function renderSales() {
```

```
    const tbody =
        document.getElementById("salesTableBody");
```

```
    if (!tbody) return;
```

```
    tbody.innerHTML = "";
```

```
    db.sales.forEach(sale => {
```

```
        tbody.innerHTML += `
```

```
        <tr>
```

```
            <td>${escapeHTML(sale.invoice)}</td>
```

```
            <td>${escapeHTML(sale.customerName)}</td>
```

```
            <td>${escapeHTML(sale.productName)}</td>
```

```
            <td>${sale.quantity}</td>
```

```
            <td>₹${formatNumber(sale.total)}</td>
```

```
<td>${escapeHTML(sale.date)}</td>
```

```
<td><button class="btn btn-danger btn-small"
onclick="removeSale(${sale.id})">Remove</button></td>
```

```
</tr>
```

```
`;
```

```
});
```

```
}
```

```
/* =====
```

```
ACCOUNTING
```

```
===== */
```

```
function addTransaction() {
```

```
    const description =
```

```
        document.getElementById(
```

```
            "transactionDescription"
```

```
        ).value.trim();
```

```
    const type =
```

```
        document.getElementById(
```

```
            "transactionType"
```

```
        ).value;
```

```
    const amount =
```

```
Number(  
  document.getElementById(  
    "transactionAmount"  
  ).value  
);  
  
if (!description || !amount) {  
  
  showToast(  
    "Enter transaction details."  
  );  
  
  return;  
}  
  
db.transactions.push({  
  
  id: getNextId(db.transactions),  
  
  description,  
  
  type,  
  
  amount,  
  
  date: new Date().toLocaleString()  
  
});
```

```
saveDB();
```

```
addAudit(  
    getCurrentUser(),  
    "Transaction Added",  
    description  
);
```

```
document.getElementById(  
    "transactionDescription"  
).value = "";
```

```
document.getElementById(  
    "transactionAmount"  
).value = "";
```

```
renderTransactions();
```

```
refreshReports();
```

```
showToast(  
    "Transaction added successfully."  
);
```

```
}
```

```
function renderTransactions() {

  const tbody =
    document.getElementById(
      "transactionTableBody"
    );

  if (!tbody) return;

  tbody.innerHTML = "";

  db.transactions.forEach(transaction => {

    const badge =
      transaction.type === "Income"
      ? "badge-success"
      : "badge-danger";

    tbody.innerHTML += `

    <tr>

      <td>${transaction.id}</td>

      <td>${escapeHTML(transaction.description)}</td>

      <td>
        <span class="badge ${badge}">
          ${transaction.type}
        </span>
      </td>
    </tr>
  `;
  }
```

```

        </span>
    </td>

    <td>₹${formatNumber(transaction.amount)}</td>

    <td>${escapeHTML(transaction.date)}</td>

    <td><button class="btn btn-danger btn-small"
onclick="removeTransaction(${transaction.id})">Remove</button></td>

</tr>

`;

});

}

/* =====
PROCUREMENT
===== */

function addPurchase() {

    const supplier =
        document.getElementById(
            "purchaseSupplier"
        ).value.trim();

    const product =

```

```
document.getElementById(  
  "purchaseProduct"  
)?.value.trim();  
  
const quantity =  
  Number(  
    document.getElementById(  
      "purchaseQuantity"  
    ).value  
  );  
  
const amount =  
  Number(  
    document.getElementById(  
      "purchaseAmount"  
    ).value  
  );  
  
if (!supplier || !product || !quantity) {  
  
  showToast(  
    "Enter purchase details."  
  );  
  
  return;  
}  
  
const purchase = {
```

```
id: getNextId(db.purchases),
```

```
po:
```

```
  "PO-" +
```

```
  new Date().getFullYear() +
```

```
  "-" +
```

```
  String(
```

```
    getNextId(db.purchases)
```

```
  ).padStart(4, "0"),
```

```
supplier,
```

```
product,
```

```
quantity,
```

```
amount,
```

```
date: new Date().toLocaleString()
```

```
};
```

```
db.purchases.push(purchase);
```

```
if (amount) {
```

```
  db.transactions.push({
```

```
    id: getNextId(db.transactions),
```

description:

"Purchase " + purchase.po,

type: "Expense",

amount,

date: new Date().toLocaleString()

});

}

saveDB();

addAudit(
 currentUser(),
 "Purchase Created",
 purchase.po
);

document.getElementById(
 "purchaseSupplier"
).value = "";

document.getElementById(
 "purchaseProduct"

```
).value = "";
```

```
document.getElementById(  
  "purchaseQuantity"  
).value = "";
```

```
document.getElementById(  
  "purchaseAmount"  
).value = "";
```

```
renderPurchases();
```

```
renderTransactions();
```

```
refreshReports();
```

```
showToast(  
  "Purchase order created."  
);
```

```
}
```

```
function renderPurchases() {
```

```
  const tbody =  
    document.getElementById(  
      "purchaseTableBody"  
    );
```

```
if (!tbody) return;
```

```
tbody.innerHTML = "";
```

```
db.purchases.forEach(purchase => {
```

```
tbody.innerHTML += `
```

```
<tr>
```

```
<td>${escapeHTML(purchase.po)}</td>
```

```
<td>${escapeHTML(purchase.supplier)}</td>
```

```
<td>${escapeHTML(purchase.product)}</td>
```

```
<td>${purchase.quantity}</td>
```

```
<td>₹${formatNumber(purchase.amount || 0)}</td>
```

```
<td>${escapeHTML(purchase.date)}</td>
```

```
<td><button class="btn btn-danger btn-small"
onclick="removePurchase(${purchase.id})">Remove</button></td>
```

```
</tr>
```

```
`;
```

```

    });

}

/* =====
REPORTS
===== */

function clearPurchaseForm() {
  ["purchaseSupplier", "purchaseProduct", "purchaseQuantity", "purchaseAmount"].forEach(id => {
    const el = document.getElementById(id);
    if (el) el.value = "";
  });
  showToast("Purchase order form cleared.");
}

function removeEmployee(id) {
  const item = db.employees.find(x => x.id === id);
  if (!item || !confirm("Remove this employee?")) return;
  db.employees = db.employees.filter(x => x.id !== id);
  saveDB();
  addAudit(getCurrentUser(), "Employee Removed", item.name);
  renderEmployees(); refreshDashboard();
  showToast("Employee removed.");
}

function removeProduct(id) {
  const item = db.products.find(x => x.id === id);
  if (!item || !confirm("Remove this product?")) return;

```

```
db.products = db.products.filter(x => x.id !== id);  
saveDB();  
addAudit(getCurrentUser(), "Product Removed", item.name);  
renderProducts(); refreshDashboard(); updateSaleProductList();  
showToast("Product removed.");  
}
```

```
function removeCustomer(id) {  
  const item = db.customers.find(x => x.id === id);  
  if (!item || !confirm("Remove this customer? Existing sales will be kept.")) return;  
  db.customers = db.customers.filter(x => x.id !== id);  
  saveDB();  
  addAudit(getCurrentUser(), "Customer Removed", item.name);  
  renderCustomers(); refreshDashboard();  
  showToast("Customer removed.");  
}
```

```
function removeSale(id) {  
  const item = db.sales.find(x => x.id === id);  
  if (!item || !confirm("Remove this sale and restore its stock?")) return;  
  const product = db.products.find(x => x.id === item.productId);  
  if (product) product.stock += Number(item.quantity || 0);  
  db.sales = db.sales.filter(x => x.id !== id);  
  db.transactions = db.transactions.filter(t => t.description !== "Sale " + item.invoice);  
  saveDB();  
  addAudit(getCurrentUser(), "Sale Removed", item.invoice);  
  renderSales(); renderProducts(); renderTransactions(); refreshDashboard(); refreshReports();  
  showToast("Sale removed and stock restored.");  
}
```

```
function removeTransaction(id) {
```

```

const item = db.transactions.find(x => x.id === id);
if (!item || !confirm("Remove this accounting transaction?")) return;
db.transactions = db.transactions.filter(x => x.id !== id);
saveDB();
addAudit(getCurrentUser(), "Transaction Removed", item.description);
renderTransactions(); refreshReports(); refreshDashboard();
showToast("Transaction removed.");
}

```

```

function removePurchase(id) {
  const item = db.purchases.find(x => x.id === id);
  if (!item || !confirm("Remove this purchase order?")) return;
  db.purchases = db.purchases.filter(x => x.id !== id);
  db.transactions = db.transactions.filter(t => t.description !== "Purchase " + item.po);
  saveDB();
  addAudit(getCurrentUser(), "Purchase Removed", item.po);
  renderPurchases(); renderTransactions(); refreshReports(); refreshDashboard();
  showToast("Purchase order removed.");
}

```

```

function pdfEscape(value) {
  return String(value).replace(/\\/g, "\\").replace(/\(/g, "\\(").replace(/\)/g, "\\)").replace(/[^\x20-\x7E]/g, "?");
}

```

```

function downloadReportsPDF() {
  const income = db.transactions.filter(t => t.type === "Income").reduce((a,t) => a + Number(t.amount || 0), 0);
  const expense = db.transactions.filter(t => t.type !== "Income").reduce((a,t) => a + Number(t.amount || 0), 0);
  const balance = income - expense;
  const lines = [

```

```

"ERP SYSTEM - BUSINESS REPORT",
"Generated: " + new Date().toLocaleString(),
"",
"SUMMARY",
"Employees: " + db.employees.length,
"Customers: " + db.customers.length,
"Products: " + db.products.length,
"Sales Orders: " + db.sales.length,
"Purchase Orders: " + db.purchases.length,
"Transactions: " + db.transactions.length,
"",
"FINANCIAL SUMMARY",
"Income: INR " + income.toFixed(2),
"Expenses: INR " + expense.toFixed(2),
"Balance: INR " + balance.toFixed(2),
"",
"RECENT SALES"
];

db.sales.slice(-12).reverse().forEach(s => lines.push((s.invoice || "Sale") + " | " + (s.customerName || "") + " | INR " + Number(s.total || 0).toFixed(2)));

lines.push("", "PURCHASE ORDERS");

db.purchases.slice(-12).reverse().forEach(p => lines.push((p.po || "PO") + " | " + (p.supplier || "") + " | INR " + Number(p.amount || 0).toFixed(2)));

const perPage = 42, pages = [];
for (let i = 0; i < lines.length; i += perPage) pages.push(lines.slice(i, i + perPage));
if (!pages.length) pages.push(["ERP SYSTEM - BUSINESS REPORT"]);

const objects = [];
objects[1] = "<< /Type /Catalog /Pages 2 0 R >>";
const pageNums = [];
const firstPageObj = 4;

```

```

pages.forEach((page, pi) => {
    const pageObj = firstPageObj + pi * 2;
    const contentObj = pageObj + 1;
    pageNums.push(pageObj + " 0 R");
    let content = "BT /F1 10 Tf 40 760 Td 14 TL\n";
    page.forEach((line, li) => {
        if (li === 0) content += "/F1 14 Tf (" + pdfEscape(line) + ") Tj /F1 10 Tf T*\n";
        else content += "(" + pdfEscape(line) + ") Tj T*\n";
    });
    content += "ET";
    objects[pageObj] = "<< /Type /Page /Parent 2 0 R /MediaBox [0 0 595 842] /Resources << /Font  

    << /F1 3 0 R >> >> /Contents " + contentObj + " 0 R >>";
    objects[contentObj] = "<< /Length " + content.length + " >>\nstream\n" + content +  

    "\nendstream";
});
objects[2] = "<< /Type /Pages /Kids [" + pageNums.join(" ") + "] /Count " + pages.length + " >>";
objects[3] = "<< /Type /Font /Subtype /Type1 /BaseFont /Helvetica >>";

let pdf = "%PDF-1.4\n%\xE2\xE3\xCF\xD3\n";
const offsets = [0];
for (let i = 1; i < objects.length; i++) {
    offsets[i] = pdf.length;
    pdf += i + " 0 obj\n" + objects[i] + "\nendobj\n";
}
const xref = pdf.length;
pdf += "xref\n0 " + objects.length + "\n0000000000 65535 f\n";
for (let i = 1; i < objects.length; i++) pdf += String(offsets[i]).padStart(10, "0") + " 00000 n\n";
pdf += "trailer\n<< /Size " + objects.length + " /Root 1 0 R >>\nstartxref\n" + xref + "\n%%EOF";
const blob = new Blob([new Uint8Array([...pdf].map(c => c.charCodeAt(0)))], {type:
"application/pdf"});
const url = URL.createObjectURL(blob);

```

```
const a = document.createElement("a"); a.href = url; a.download = "ERP_Business_Report.pdf";
a.click();

setTimeout(() => URL.revokeObjectURL(url), 1000);

addAudit(getCurrentUser(), "Report Downloaded", "ERP Business Report PDF");

showToast("PDF report downloaded.");

}
```

```
function refreshReports() {
```

```
    let income = 0;
```

```
    let expense = 0;
```

```
    db.transactions.forEach(t => {
```

```
        if (t.type === "Income") {
```

```
            income += Number(t.amount);
```

```
        } else {
```

```
            expense += Number(t.amount);
```

```
        }
```

```
    });
```

```
    const balance =
```

```
        income - expense;
```

```
document.getElementById(  
    "reportIncome"  
).textContent =  
    "₹" + formatNumber(income);
```

```
document.getElementById(  
    "reportExpense"  
).textContent =  
    "₹" + formatNumber(expense);
```

```
document.getElementById(  
    "reportBalance"  
).textContent =  
    "₹" + formatNumber(balance);
```

```
document.getElementById(  
    "reportOrders"  
).textContent =  
    db.sales.length;
```

```
const salesPercent =  
    Math.min(  
        100,  
        db.sales.length * 10  
    );
```

```
const inventoryTotal =  
  db.products.reduce(  
    (sum,p) =>  
      sum + Number(p.stock || 0),  
    0  
  );
```

```
const inventoryPercent =  
  Math.min(  
    100,  
    inventoryTotal  
      ? Math.round(  
        db.products.length /  
        Math.max(inventoryTotal,1)  
        * 100  
      )  
    : 0  
  );
```

```
const customerPercent =  
  Math.min(  
    100,  
    db.customers.length * 10  
  );
```

```
document.getElementById(  
  "salesPercent"
```

```
).textContent =  
    salesPercent + "%";
```

```
document.getElementById(  
    "salesProgress"  
).style.width =  
    salesPercent + "%";
```

```
document.getElementById(  
    "inventoryPercent"  
).textContent =  
    inventoryPercent + "%";
```

```
document.getElementById(  
    "inventoryProgress"  
).style.width =  
    inventoryPercent + "%";
```

```
document.getElementById(  
    "customerPercent"  
).textContent =  
    customerPercent + "%";
```

```
document.getElementById(  
    "customerProgress"  
).style.width =
```

```

        customerPercent + "%";

    }

    /* =====
    USERS
    ===== */

function renderUsers() {

    const tbody =
        document.getElementById(
            "usersTableBody"
        );

    if (!tbody) return;

    const users = JSON.parse(
        localStorage.getItem(USERS_KEY) || "[]"
    );

    tbody.innerHTML = "";

    users.forEach(user => {

        tbody.innerHTML += `

```

```
<tr>
```

```
<td>${escapeHTML(user.name)}</td>
```

```
<td>${escapeHTML(user.email)}</td>
```

```
<td>${escapeHTML(user.username)}</td>
```

```
<td>
```

```
<span class="badge badge-info">
```

```
  ${escapeHTML(user.role)}
```

```
</span>
```

```
</td>
```

```
</tr>
```

```
`;
```

```
});
```

```
}
```

```
/* =====
```

```
AUDIT LOGS
```

```
===== */
```

```
function addAudit(user, action, details) {
```

```
  if (!db) return;
```

```
db.audit.unshift({  
  
  date: new Date().toLocaleString(),  
  
  user: user || "System",  
  
  action,  
  
  details  
  
});
```

```
if (db.audit.length > 500) {
```

```
  db.audit =  
    db.audit.slice(0,500);
```

```
}
```

```
saveDB();
```

```
}
```

```
function renderAudit() {
```

```
  const tbody =  
    document.getElementById(
```

```
        "auditTableBody"
    );

    if (!tbody) return;

    tbody.innerHTML = "";

    db.audit.forEach(log => {

        tbody.innerHTML += `

        <tr>

            <td>${escapeHTML(log.date)}</td>

            <td>${escapeHTML(log.user)}</td>

            <td>${escapeHTML(log.action)}</td>

            <td>${escapeHTML(log.details)}</td>

        </tr>

        `;

    });

}
```

```

function clearAudit() {

    if (
        !confirm(
            "Are you sure you want to clear all audit logs?"
        )
    ){
        return;
    }

    db.audit = [];

    saveDB();

    renderAudit();

    showToast(
        "Audit logs cleared."
    );

}

/* =====
DASHBOARD
===== */

function refreshDashboard() {

```

```
document.getElementById(  
  "employeeCount"  
).textContent =  
  db.employees.length;
```

```
document.getElementById(  
  "productCount"  
).textContent =  
  db.products.length;
```

```
document.getElementById(  
  "customerCount"  
).textContent =  
  db.customers.length;
```

```
const totalSales =  
  db.sales.reduce(  
    (sum,sale) =>  
      sum + Number(sale.total || 0),  
    0  
  );
```

```
document.getElementById(  
  "salesTotal"  
).textContent =  
  "₹" + formatNumber(totalSales);
```

```
const tbody =
  document.getElementById(
    "recentSalesTable"
  );

tbody.innerHTML = "";

db.sales.slice(0,8).forEach(sale => {

  tbody.innerHTML += `

    <tr>

      <td>${escapeHTML(sale.invoice)}</td>

      <td>${escapeHTML(sale.customerName)}</td>

      <td>₹${formatNumber(sale.total)}</td>

      <td>${escapeHTML(sale.date)}</td>

    </tr>

  `;

});

}
```

```
/* =====  
SEARCH  
===== */
```

```
function searchTable(input, tableId) {  
  
    const filter =  
        input.value.toLowerCase();  
  
    const table =  
        document.getElementById(tableId);  
  
    if (!table) return;  
  
    const rows =  
        table.getElementsByTagName("tr");  
  
    for (  
        let i = 1;  
        i < rows.length;  
        i++  
    ) {  
  
        const text =  
            rows[i].textContent.toLowerCase();
```

```

        rows[i].style.display =
            text.includes(filter)
                ? ""
                : "none";

    }

}

/* =====
UTILITIES
===== */

function getNextId(array) {

    if (!array.length) {
        return 1;
    }

    return (
        Math.max(
            ...array.map(
                item => Number(item.id) || 0
            )
        ) + 1
    );

}

```

```
function formatNumber(number) {  
  
    return Number(number || 0)  
        .toLocaleString("en-IN");  
  
}
```

```
function escapeHTML(value) {  
  
    return String(value ?? "")  
        .replace(/&/g, "&amp;")  
        .replace(/</g, "&lt;")  
        .replace(/>/g, "&gt;")  
        .replace(/"/g, "&quot;")  
        .replace(/'/g, "&#039;");  
  
}
```

```
function getCurrentUser() {  
  
    const user =  
        JSON.parse(  
            sessionStorage.getItem(  
                SESSION_KEY  
            ) || "null"  
        );  
  
    return user
```

```
    ? user.username  
    : "System";  
  
}
```

```
function showToast(message) {  
  
    const toast =  
        document.getElementById("toast");  
  
    toast.textContent = message;  
  
    toast.classList.add("show");  
  
    setTimeout(  
        () => toast.classList.remove("show"),  
        2500  
    );  
  
}
```

```
/* =====  
REFRESH EVERYTHING  
===== */
```

```
function refreshAll() {  
  
    initializeDatabase();
```

```
renderEmployees();
```

```
renderProducts();
```

```
renderCustomers();
```

```
renderSales();
```

```
renderTransactions();
```

```
renderPurchases();
```

```
renderUsers();
```

```
renderAudit();
```

```
refreshDashboard();
```

```
refreshReports();
```

```
}
```

```
/* =====
```

```
ENTER KEY LOGIN
```

```
===== */
```

```
document.addEventListener(
```

```
    "keydown",
```

```
    function(event) {
```

```

    if (
        event.key === "Enter" &&
        document.getElementById(
            "loginScreen"
        ).style.display !== "none"
    ) {

        const loginForm =
            document.getElementById(
                "loginForm"
            );

        if (
            !loginForm.classList.contains(
                "hidden"
            )
        ) {

            login();

        }

    }

}

);

/* =====
INITIALIZE

```

```
===== */

initializeTheme();

initializeDatabase();

initializeUsers();


const existingSession =
  JSON.parse(
    sessionStorage.getItem(
      SESSION_KEY
    ) || "null"
  );

if (existingSession) {

  openApplication(
    existingSession
  );

}

</script>

</body>

</html>
```